

Route Efficiency and Profitability Analysis of Nassau Candy Distribution Using Interactive Data Visualization

Author: Jason Kunene - Project 1 - 2026

Links:

GitHub: [REPO](#)

Deployed Dashboard: [Streamlit](#)

ABSTRACT

This study analyzes 4,130 cleaned transaction records from Nassau Candy Company to evaluate sales performance, factory profitability, and logistics efficiency. Using Python, Pandas, and a Streamlit dashboard with Plotly visualizations, we identify key drivers of profit and recommend route and production optimizations. Total sales analyzed were \$57,064.50 with \$37,724.68 gross profit (66.1% margin). Results show Chocolate division dominance (89.4% sales), factory concentration in two plants (95.6% profit), and no correlation between lead time and profit ($r=0.03$).

1. INTRODUCTION

Nassau Candy operates across four US regions (Pacific, Atlantic, Interior, Gulf) with multiple factories and three divisions: Chocolate, Sugar, and Other. With rising logistics costs and average lead times of 120.6 days, understanding which divisions and factories maximize gross profit is critical for route optimization.

2. PROBLEM STATEMENT

Which Division, Factory, Region, and Ship Mode maximize profit while minimizing lead time? The objective is to build an interactive dashboard that allows stakeholders to filter and visualize performance.

3. METHODOLOGY

3.1 Data Source: Raw sales data (>5k rows) cleaned to 4,130 rows.

3.2 Tools: Python (Pandas), Plotly Express, Streamlit, GitHub.

3.3 Cleaning Steps: Removed nulls and duplicates, parsed Order Date and Ship

Date to datetime, created $\text{Lead_Time_Days} = \text{Ship Date} - \text{Order Date}$, standardized Division/Region names, validated Sales, Units, Gross Profit, Cost.

4. RESULTS

- Overall KPIs: 4,130 Orders | \$57,064.50 Sales | \$37,724.68 Profit | Avg 120.6 days lead | \$9.13 profit per order.
- By Division: Chocolate \$51,021 (89.4%), Other \$3,980 (7.0%), Sugar \$2,063 (3.6%).
- By Factory: Lot's O' Nuts \$21,847 profit (57.9%), Wicked Choccy's \$14,220 (37.7%), Horchow \$1,105, Sweet Addition \$552. Top 2 factories = 95.6% of profit.
- By Region: Pacific \$18,604 (32.6%), Atlantic \$16,235 (28.5%), Interior \$13,809 (24.2%), Gulf \$8,415 (14.8%).
- Logistics: Ship Mode - Standard Class 89% of orders, Second Class 6%, First Class 3%, Same Day 2%. Correlation Lead_Time vs Profit $r=0.03$ (no relationship). Profit stable across 0-365 days.

5. DISCUSSION

Production efficiency matters more than shipping speed. Chocolate dominance suggests product-market fit. Factory concentration indicates economies of scale at Lot's O' Nuts. Lack of correlation suggests customers are not paying a premium for faster shipping; Standard Class is optimal.

6. CONCLUSION AND RECOMMENDATION

1. Prioritize Chocolate Division - discontinue low-performing Sugar SKUs.
2. Shift production to Lot's O' Nuts and Wicked Choccy's.
3. Maintain Standard Class shipping - no ROI on faster modes.
4. Optimize Pacific and Atlantic routes - highest demand.
5. Focus on factory efficiency, not shipping speed.

7. DASHBOARD IMPLEMENTATION

Built with Streamlit + Plotly: KPI cards, interactive Division/Region filters, 4 charts (Sales by Division bar, Profit by Factory bar, Lead Time vs Profit scatter colored by Ship Mode, Sales by Region pie), searchable data table. Auto-loads CSV from GitHub. Deployed on Streamlit Cloud.